By (D.1) and (D.2) we also obtain

$$\begin{aligned}\frac{d\lambda_U}{d\pi_{2n+1}} &= \left(\mathcal{S}_{2(n+1)}\,\mathcal{S}^{\circ}_{2n,2p}\,\mathcal{S}_{2p-1}\right)\left(\frac{d\lambda_U}{d\pi_{2p-1}}\right) \quad \text{and}\\ \frac{d\nu_V}{d\pi_{2n}} &= \left(\mathcal{S}_{2n+1}\,\mathcal{S}^{\circ}_{2n-1,2p-1}\,\mathcal{S}_{2(p-1)}\right)\left(\frac{d\nu_V}{d\pi_{2(p-1)}}\right).\end{aligned} \tag{D.4}$$

D.2. **Some quantitative estimates.**

Proposition D.1. *When $\jmath(\mathcal{Q})>0$, for any $n\geqslant 0$ we have the uniform estimates*

$$\|\frac{d\pi_{2n}}{d\nu_V}\|\vee\|\frac{d\nu_V}{d\pi_{2n}}\|\leqslant \hbar(\mathcal{Q})^{-2} \quad \textit{and} \quad \|\frac{d\pi_{2n+1}}{d\lambda_U}\|\vee\|\frac{d\lambda_U}{d\pi_{2n+1}}\|\leqslant \hbar(\mathcal{Q})^{-2}. \tag{D.5}$$

Proof. Recalling that $\jmath(\mathcal{S}_n)\wedge\jmath(\mathcal{Q})\geqslant\hbar(\mathcal{S}_n)=\hbar(\mathcal{Q})$, for any given $x\in\mathbb{X}$ we have

$$\hbar(\mathcal{Q})\;\mathcal{S}_{2n}(x,dy_2)\leqslant\mathcal{S}^{\circ}_{2n}(y_1,dy_2)=(\mathcal{S}_{2n-1}\mathcal{S}_{2n})(y_1,dy_2)\leqslant\frac{1}{\hbar(\mathcal{Q})}\;\mathcal{S}_{2n}(x,dy_2).$$

This yields, for any $(y_1,\overline{y}_1)\in\mathbb{Y}^2$, the estimate

$$\hbar(\mathcal{Q})^2\;\mathcal{S}^{\circ}_{2n}(\overline{y}_1,dy_2)\leqslant\mathcal{S}^{\circ}_{2n}(y_1,dy_2)\leqslant\frac{1}{\hbar(\mathcal{Q})^2}\;\mathcal{S}^{\circ}_{2n}(\overline{y}_1,dy_2)$$

and integrating w.r.t. $\nu_V(d\overline{y}_1)$ and $\pi_{2(n-1)}(dy_1)$ we find that

$$\hbar(\mathcal{Q})^2\;\nu_V(dy_2)\leqslant\pi_{2n}(dy_2)\leqslant\frac{1}{\hbar(\mathcal{Q})^2}\;\nu_V(dy_2).$$

Similarly, integrating w.r.t. $\nu_V(dy_1)$ and $\pi_{2(n-1)}(d\overline{y}_1)$ we also obtain

$$\hbar(\mathcal{Q})^2\;\pi_{2n}(dy_2)\leqslant\nu_V(dy_2)\leqslant\frac{1}{\hbar(\mathcal{Q})^2}\;\pi_{2n}(dy_2).$$

From the inequalities above we conclude that

$$\frac{d\pi_{2n}}{d\nu_V}\vee\frac{d\nu_V}{d\pi_{2n}}\leqslant\hbar(\mathcal{Q})^{-2} \quad\text{and}\quad \frac{d\pi_{2n+1}}{d\lambda_U}\vee\frac{d\lambda_U}{d\pi_{2n+1}}\leqslant\hbar(\mathcal{Q})^{-2}.$$

■

More generally these Radon Nikodym functions are unbounded and we need to carry a more refined analysis.

Proposition D.2. *If Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S}^{\circ})$ holds for some $n_0\geqslant 1$ then there exists $c_\epsilon>0$ such that*

$$\sup_{n\geqslant n_0}\|\frac{d\pi_{2n}}{d\nu_V}\|_\psi\leqslant c_\epsilon\;\|\frac{d\pi_{2(n_0-1)}}{d\nu_V}\|_\psi \quad\text{and}\quad \sup_{n\geqslant n_0}\|\frac{d\pi_{2n+1}}{d\lambda_U}\|_\varphi\leqslant c_\epsilon\;\|\frac{d\pi_{2n_0-1}}{d\lambda_U}\|_\varphi. \tag{D.6}$$

Proof. Using (D.3) and (4.6), we see that, for any $n\geqslant n_0$,

$$\begin{aligned}\frac{d\pi_{2n}}{d\nu_V} &\leqslant \|\frac{d\pi_{2(n_0-1)}}{d\nu_V}\|_\psi\;\left(\mathcal{S}^{\circ}_{2n}\ldots\mathcal{S}^{\circ}_{2n_0}\right)(\psi)\\ &\leqslant \|\frac{d\pi_{2(n_0-1)}}{d\nu_V}\|_\psi\left(\epsilon^{n-n_0}\psi+\frac{c}{1-\epsilon}\right)\leqslant c_\epsilon\;\|\frac{d\pi_{2(n_0-1)}}{d\nu_V}\|_\psi\;\psi, \quad \text{with } c_\epsilon:=\left(1+\frac{c}{1-\epsilon}\right).\end{aligned}$$

This implies that

$$\sup_{n\geqslant n_0}\|\frac{d\pi_{2n}}{d\nu_V}\|_\psi\leqslant c_\epsilon\;\|\frac{d\pi_{2(n_0-1)}}{d\nu_V}\|_\psi \quad\text{and, similarly,}\quad \sup_{n\geqslant n_0}\|\frac{d\pi_{2n+1}}{d\lambda_U}\|_\varphi\leqslant c_\epsilon\;\|\frac{d\pi_{2n_0-1}}{d\lambda_U}\|_\varphi.$$

■

Proposition D.3. *In addition, when Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S})$ holds for some $n_0 \geqslant 0$, there exists $c_\epsilon > 0$ such that*

$$\sup_{n\geqslant n_0}\left(\|\frac{d\pi_{2n+1}}{d\lambda_U}\|_\varphi \vee \|\frac{d\lambda_U}{d\pi_{2n+1}}\|_\varphi\right) \leqslant c_\epsilon \left(\|\frac{d\pi_{2n_0+1}}{d\lambda_U}\|_\varphi \vee \|\frac{d\lambda_U}{d\pi_{2n_0+1}}\|_\varphi\right),$$

as well as

$$\sup_{n\geqslant n_0}\left(\|\frac{d\pi_{2n}}{d\nu_V}\|_\psi \vee \|\frac{d\nu_V}{d\pi_{2n}}\|_\psi\right) \leqslant c_\epsilon \left(\|\frac{d\pi_{2n_0}}{d\nu_V}\|_\psi \vee \|\frac{d\nu_V}{d\pi_{2n_0}}\|_\psi\right).$$

Proof. If Condition $\mathcal{D}_{\varphi,\psi}(\mathcal{S})$ is satisfied for some integer $n_0 \geqslant 0$ then, by Theorem 4.6, the drift condition (4.6) also holds for any $n > n_0$, which implies that

$$\sup_{n\geqslant n_0}\|\frac{d\pi_{2n}}{d\nu_V}\|_\psi \leqslant c_\epsilon\,\|\frac{d\pi_{2n_0}}{d\nu_V}\|_\psi \quad \text{and, similarly,} \quad \sup_{n\geqslant n_0}\|\frac{d\pi_{2n+1}}{d\lambda_U}\|_\varphi \leqslant c_\epsilon\,\|\frac{d\pi_{2n_0+1}}{d\lambda_U}\|_\varphi.$$

Arguing as in the proof of Proposition D.2, now combining the drift conditions (4.13) with (D.4), we find that

$$\sup_{n\geqslant n_0}\|\frac{d\nu_V}{d\pi_{2n}}\|_\psi \leqslant c_\epsilon\,\|\frac{d\nu_V}{d\pi_{2n_0}}\|_\psi \quad \text{and} \quad \sup_{n\geqslant n_0}\|\frac{d\lambda_U}{d\pi_{2n+1}}\|_\varphi \leqslant c_\epsilon\,\|\frac{d\lambda_U}{d\pi_{2n_0+1}}\|_\varphi.$$

■

Under the stronger Condition $\mathcal{H}_\delta(U,V)$, more refined estimates can be obtained. With that end, consider the functions

$$\mathcal{W}_{V,U}(x) := \log \mathcal{Q}_V(e^{W_U})(x) \quad \text{and} \quad \mathcal{W}^{U,V}(y) := \log \mathcal{R}_U(e^{W^V})(y), \tag{D.7}$$

with the integrated cost functions (W_U, W^V) defined in (2.14) and the integral operators

$$\mathcal{Q}_V(x,dy) = e^{-W(x,y)}\,\nu_V(dy) \quad \text{and} \quad \mathcal{R}_U(y,dx) = e^{-W(x,y)}\,\lambda_U(dx).$$

The next technical lemma is pivotal.

Lemma D.4. *For any $n \geqslant 0$ we have*

$$\begin{aligned} \log \mathcal{Q}(\exp(-V_{2n})) &\geqslant -W^V + \nu_V(V) \quad \textit{and} \\ -\nu_V(V) + \mathcal{W}^{U,V} \geqslant \log \mathcal{R}(\exp(-U_{2n})) &\geqslant -W_U, \end{aligned} \tag{D.8}$$

as well as

$$U_{2n} \geqslant U - W^V + \nu_V(V).$$

In addition, for any $n \geqslant 1$ we have

$$\mathcal{W}_{V,U} \geqslant \log \mathcal{Q}(\exp(-V_{2n})) \geqslant -W^V + \nu_V(V) \tag{D.9}$$

as well as

$$\begin{aligned} U - W^V + \nu_V(V) &\leqslant U_{2n} \leqslant U + \mathcal{W}_{V,U} \quad \textit{and} \\ V - W_U &\leqslant V_{2n} \leqslant V + \mathcal{W}^{U,V} - \nu_V(V). \end{aligned} \tag{D.10}$$

Proof. Formula (2.7) yields the following monotone properties

$$\begin{aligned} \nu_V(V_{2(n+1)}) &\leqslant \nu_V(V_{2n}) = \nu_V(V_{2(n+1)}) + \mathrm{Ent}(\nu_V \mid \pi_{2n}) \leqslant \nu_V(V_0) = 0, \\ \lambda_U(U_{2(n+1)}) &\leqslant \lambda_U(U_{2n}) = \lambda_U(U_{2(n+1)}) + \mathrm{Ent}(\lambda_U \mid \pi_{2n+1}) \leqslant \lambda_U(U_0). \end{aligned} \tag{D.11}$$

Applying Jensen's inequality, for any $n \geqslant 0$ we have

$$\begin{aligned}\log \mathcal{Q}(\exp(-V_{2n}))(x) &= \log \int \nu_V(dy)\ \exp(-W(x,y)+V(y)-V_{2n}(y)) \\ &\geqslant -W^V(x)+\nu_V(V)-\nu_V(V_{2n}) \geqslant -W^V(x)+\nu_V(V).\end{aligned}$$

By (D.11) we also have $\lambda_U(U) \geqslant \lambda_U(U_{2n})$ and, therefore, for any $n \geqslant 0$ we obtain

$$\begin{aligned}\log \mathcal{R}(\exp(-U_{2n}))(y) &= \log \int \lambda_U(dx)\ \exp(-W(x,y)+U(x)-U_{2n}(x)) \\ &\geqslant -W_U(y)+\lambda_U(U)-\lambda_U(U_{2n}) \geqslant -W_U(y)\end{aligned}$$

using Jensen's inequality again. This ends the proof of (D.8).

The integral equations in (2.4) imply that for any $n \geqslant 0$ we have

$$U_{2n} \geqslant U - W^V + \nu_V(V) \quad \text{and} \quad V_{2(n+1)} \geqslant V - W_U.$$

Using again (2.4) we see that

$$\begin{aligned}U_{2(n+1)} - U = \log \mathcal{Q}(e^{-V_{2(n+1)}}) &\leqslant \log \mathcal{Q}_V(e^{W_U}) = \mathcal{W}_{V,U} \quad \text{and} \\ V_{2(n+1)} - V = \log \mathcal{R}(e^{-U_{2n}}) &\leqslant -\nu_V(V) + \log \mathcal{R}_U(e^{W^V}) = -\nu_V(V) + \mathcal{W}^{U,V}.\end{aligned}$$

■

Proposition D.5. *If Condition $\mathcal{H}_\delta(U,V)$ is satisfied for some $\delta \in]0,1]$, then*

$$\|e^{\mathcal{W}^{U,V}}\|_{\mathbb{L}_\infty(\nu)} \vee \|e^{\mathcal{W}_{V,U}}\|_{\mathbb{L}_\infty(\lambda)} < \infty. \tag{D.12}$$

In addition, the exists some constant $c > 0$ such that, for any $n \geqslant 1$,

$$\frac{d\pi_{2n}}{d\nu_V} \vee \frac{d\nu_V}{d\pi_{2n}} \leqslant c\ e^{W_U} \quad \text{and} \quad \frac{d\pi_{2n+1}}{d\lambda_U} \vee \frac{d\lambda_U}{d\pi_{2n+1}} \leqslant c\ e^{W^V}. \tag{D.13}$$

Proof. Since U is locally bounded with compact sub-level sets for any $\delta \geqslant 0$ and any sufficiently large $r > 0$ we have

$$e^{-\delta U} = e^{-\delta U}\ 1_{\{U>r\}} + e^{-\delta U}\ 1_{\{U\leqslant r\}} \leqslant \varsigma_\delta^U(r) \quad \text{and} \quad \|e^{-\delta U}\|^{-1}_{\mathbb{L}_\infty(\lambda)} \geqslant 1/\varsigma_\delta^U(r)$$

with the parameters

$$\varsigma_\delta^U(r) := e^{-\delta r} + e^{\delta c_U(r)} \quad \text{with} \quad c_U(r) := \sup_{x\in C_U(r)} |U(x)| \quad \text{and} \quad C_U(r) = \{x\ :\ U(x) \leqslant r\}$$

This yields

$$\lambda_{(1-\delta)U}\left(e^{W^V}\right) = \lambda_U\left(e^{W^V}\ e^{\delta U}\right) \geqslant \|e^{-\delta U}\|^{-1}_{\mathbb{L}_\infty(\lambda)}\ \lambda_U\left(e^{W^V}\right).$$

In the same vein, we obtain

$$\nu_{(1-\delta)V}\left(e^{W_U}\right) = \nu_V\left(e^{W_U}\ e^{\delta V}\right) \geqslant \|e^{-\delta V}\|^{-1}_{\mathbb{L}_\infty(\nu)}\ \nu_V\left(e^{W_U}\right)$$

and conclude that

$$\begin{aligned}\|e^{\mathcal{W}_{V,U}}\|_{\mathbb{L}_\infty(\lambda)} &= \|\mathcal{Q}_V(e^{W_U})\|_{\mathbb{L}_\infty(\lambda)} \\ &\leqslant \|e^{-W}\|_{\mathbb{L}_\infty(\lambda\otimes\nu)}\ \nu_V\left(e^{W_U}\right) \\ &\leqslant \|e^{-\delta V}\|_{\mathbb{L}_\infty(\nu)}\ \|e^{-W}\|_{\mathbb{L}_\infty(\lambda\otimes\nu)}\ \nu_{(1-\delta)V}\left(e^{W_U}\right)\end{aligned}$$

as well as

$$\|e^{\mathcal{W}^{U,V}}\|_{\mathbb{L}_\infty(\nu)} = \|\mathcal{R}_U(e^{W^V})\| \leqslant \|e^{-\delta U}\|_{\mathbb{L}_\infty(\lambda)}\ \|e^{-W}\|_{\mathbb{L}_\infty(\lambda\otimes\nu)}\ \lambda_{(1-\delta)U}\left(e^{W^V}\right),$$